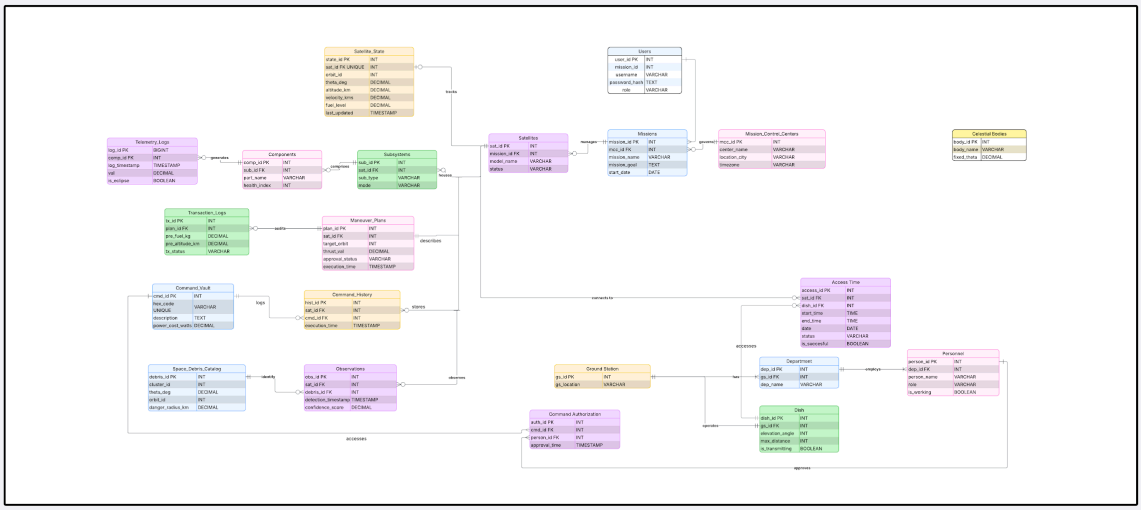
Satellite_State
state_id PK INT
sat_id FK UNIQUE INT
snsr_id INT
theta_deg DECIMAL
altitude_km DECIMAL
velocity_kms DECIMAL
fuel_level DECIMAL
last_updated TIMESTAMP
tracks
Users
user_id PK INT
mission_id INT
username VARCHAR
password_hash TEXT
role VARCHAR
Telemetry_Logs
log_id PK BIGINT
comp_id FK INT
log_timestamp TIMESTAMP
val DECIMAL
is_eclipse BOOLEAN
generates
Components
comp_id PK INT
sub_id FK INT
part_name VARCHAR
health_index INT
comprises
Subsystems
sub_id PK INT
sat_id FK INT
sub_type VARCHAR
status VARCHAR
monitors
Satellites
sat_id PK INT
mission_id FK INT
model_name VARCHAR
status VARCHAR
manages
Missions
mission_id PK INT
msc_id FK INT
mission_name VARCHAR
mission_goal TEXT
start_date DATE
governed by
Mission_Control_Centers
msc_id PK INT
center_name VARCHAR
location_city VARCHAR
timezone VARCHAR
Celestial Bodies
body_id PK INT
body_name VARCHAR
fixed_mute DECIMAL
Transaction_Logs
tx_id PK INT
plan_id FK INT
pre_fuel_kg DECIMAL
pre_altitude_km DECIMAL
tx_status VARCHAR
audits
Maneuver_Plans
plan_id PK INT
sat_id FK INT
target_orbit INT
thrust_val DECIMAL
approval_status VARCHAR
execution_time TIMESTAMP
describes
Command_Vault
cmd_id PK INT
hex_code VARCHAR UNIQUE
description TEXT
power_cost_watts DECIMAL
logs
Command_History
hist_id PK INT
sat_id FK INT
cmd_id FK INT
execution_time TIMESTAMP
stores
commands to
Access Time
access_id PK INT
sat_id FK INT
stat_id FK INT
start_time TIME
end_time TIME
date DATE
status VARCHAR
is_successful BOOLEAN
accesses
Space_Debris_Catalog
debris_id PK INT
cluster_id INT
theta_deg DECIMAL
orbit_id INT
danger_radius_km DECIMAL
identify
Observations
obs_id PK INT
sat_id FK INT
debris_id FK INT
detection_timestamp TIMESTAMP
confidence_score DECIMAL
observes
Ground Station
gs_id PK INT
gs_location VARCHAR
has
Departments
dep_id PK INT
gs_id FK INT
dep_name VARCHAR
employs
Personnel
person_id PK INT
dep_id FK INT
person_name VARCHAR
role VARCHAR
is_working BOOLEAN
operates
Dish
dish_id PK INT
gs_id FK INT
elevation_angle INT
max_distance INT
is_transmitting BOOLEAN
Command Authorization
auth_id PK INT
cmd_id FK INT
person_id FK INT
approval_time TIMESTAMP
accesses
approves